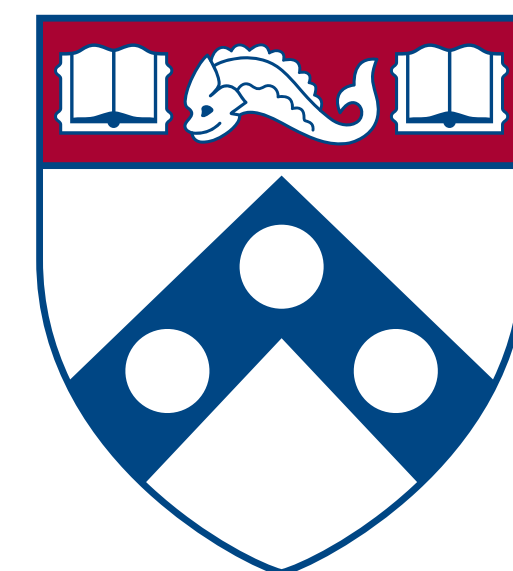


Journal Club: Alcohol Consumption in Older Adults with Health and Socioeconomic Comorbidities

Alfredo Lucas, PhD

Psych 200 Clerkship - CL Service

11/26/2024



Perelman
SCHOOL OF MEDICINE

UNIVERSITY *of* PENNSYLVANIA

Alcohol consumption is widespread

Alcohol remains a leading cause of morbidity and mortality worldwide

- Alcohol accounts for approximately 5.1% of disease burden and 5.3% of all deaths worldwide (WHO 2018)

Alcohol consumption per person, 2019

Estimated consumption of alcohol is measured in liters of pure alcohol per person aged 15 or older, per year.

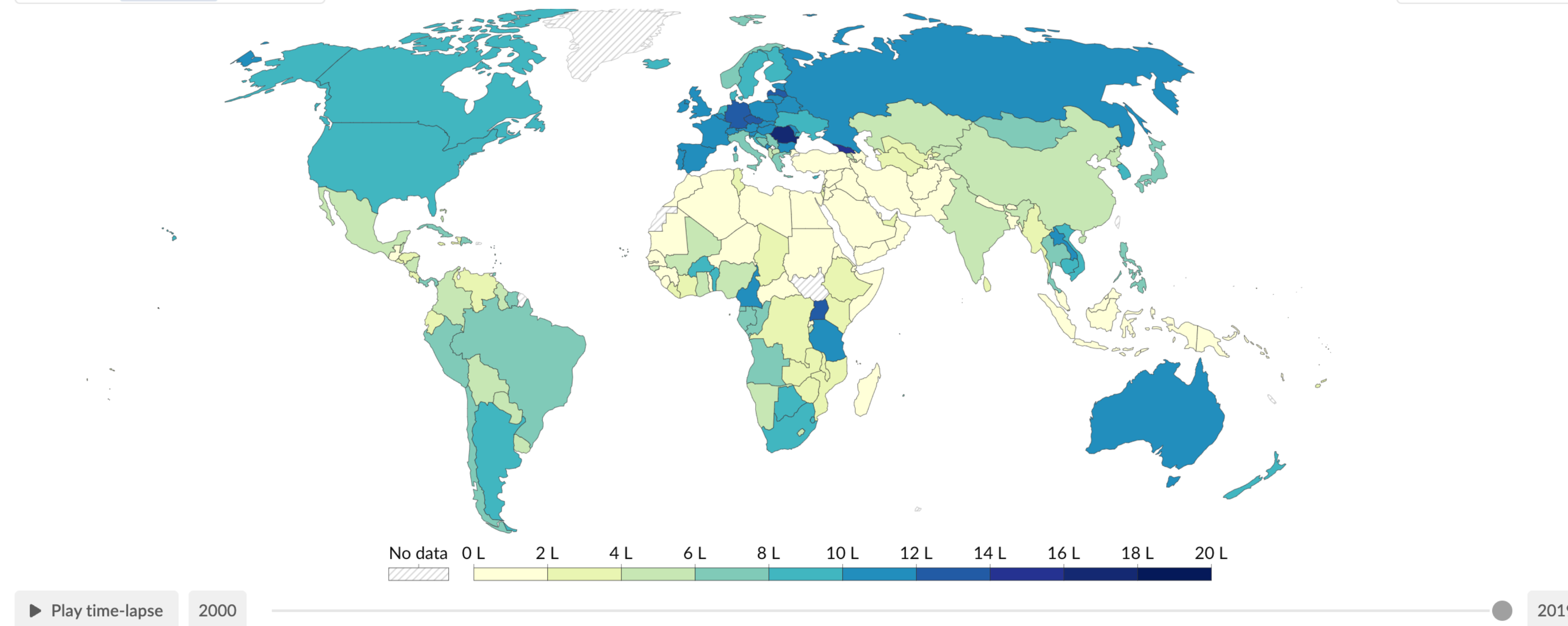
Our World
in Data

Table

Map

Chart

World



Data source: Multiple sources compiled by World Bank (2024) – [Learn more about this data](#)

OurWorldinData.org/alcohol-consumption | CC BY

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<https://ourworldindata.org/grapher/total-alcohol-consumption-per-capita-litres-of-pure-alcohol>

Alcohol consumption is widespread

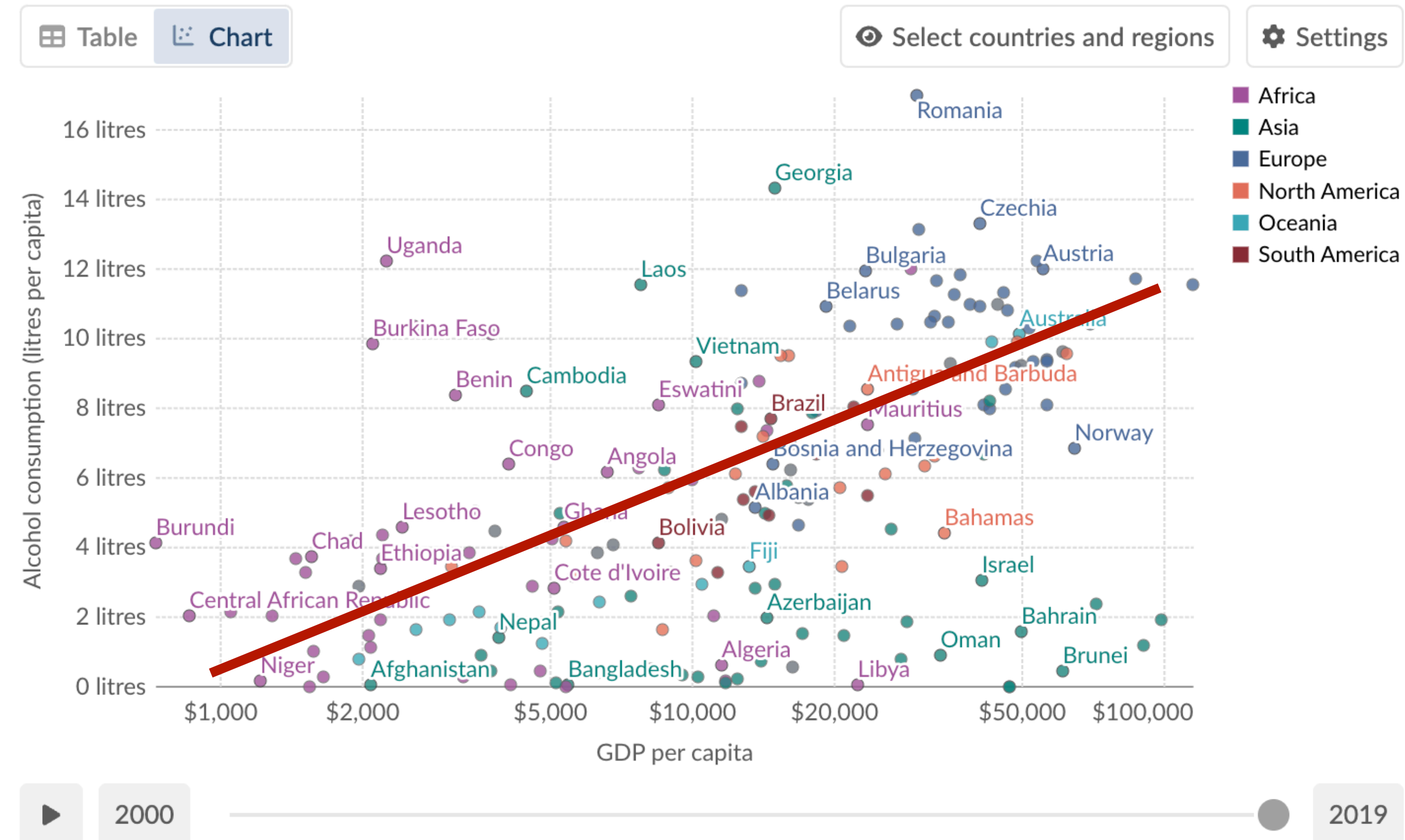
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Alcohol consumption vs. GDP per capita, 2019

Average annual alcohol consumption measured in liters of pure alcohol per person aged 15 years or older.

Our World
in Data



<https://ourworldindata.org/grapher/total-alcohol-consumption-per-capita-litres-of-pure-alcohol>

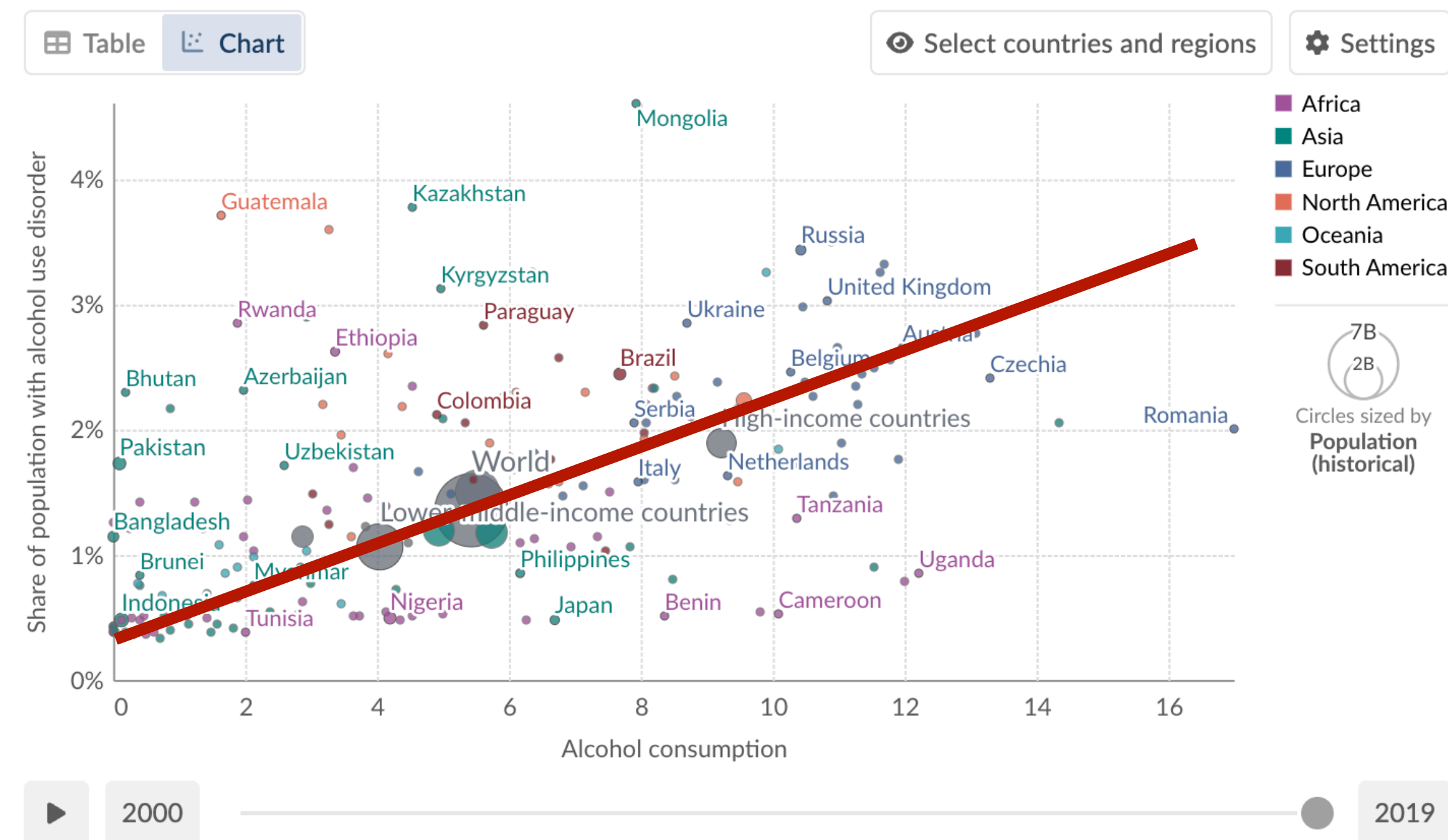
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Share of population with alcohol use disorder vs. alcohol consumption, 2019

Age-standardized prevalence of alcohol use disorder (classified based on alcohol dependency criteria), versus the average annual alcohol consumption of those aged 15+ years, measured in litres of pure alcohol per capita.



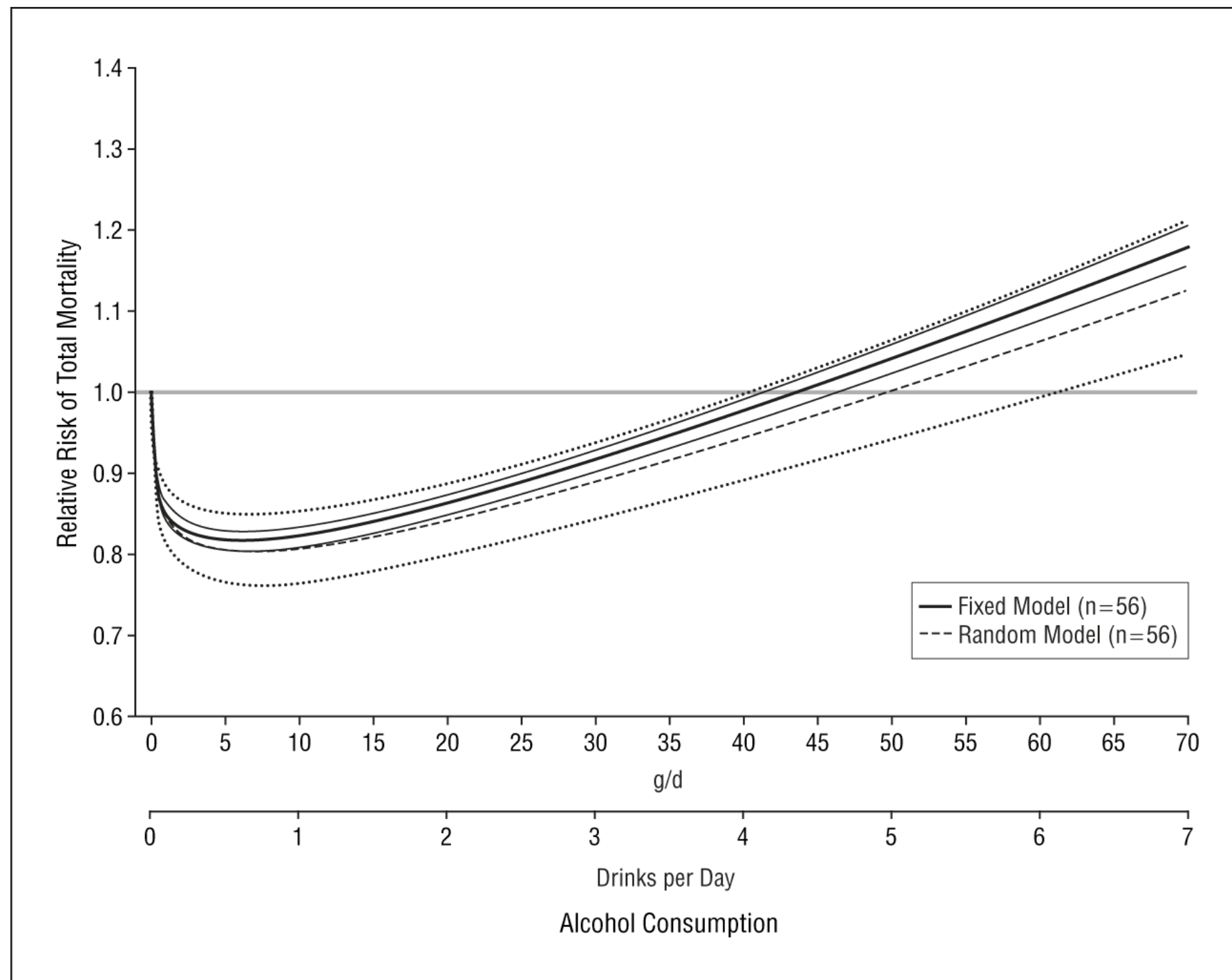
Data source: IHME, Global Burden of Disease (2024); Multiple sources compiled by World Bank (2024) – [Learn more about this data](#)

OurWorldinData.org/alcohol-consumption | CC BY

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A glass of wine with dinner is good for your heart, right?

Meta-analysis: Relative risk of total mortality as a function of alcohol consumption (~1M subjects)



“Our findings, while confirming the hazards of excess drinking, indicate potential windows of alcohol intake that may confer a net beneficial effect of moderate drinking, at least in terms of survival.”

Di Castelnuovo, JAMA Internal Medicine, 2006

- The main criticism towards many of these studies is that they suffer from significant selection bias (Niami et. al. 2017)
- One of such biases is the **abstainer bias** where:
 - The *apparent* lower mortality of light drinkers compared with abstainers could be explained by the higher death risk of the abstainers because they include former drinkers who quit alcohol due to poor health
 - Or because lifelong abstainers happen to have pre-existing health conditions that prevent them from drinking in the first place (Ng and Shelton, 2012)



Original Investigation | Public Health

Alcohol Consumption Patterns and Mortality Among Older Adults With Health-Related or Socioeconomic Risk Factors

Rosario Ortolá, MD, PhD; Mercedes Sotos-Prieto, PhD; Esther García-Esquinas, PhD; Iñaki Galán, PhD; Fernando Rodríguez-Artalejo, PhD

Research Question: *Do health-related or socioeconomic risk factors modify the associations of alcohol consumption patterns with mortality among older drinkers?*

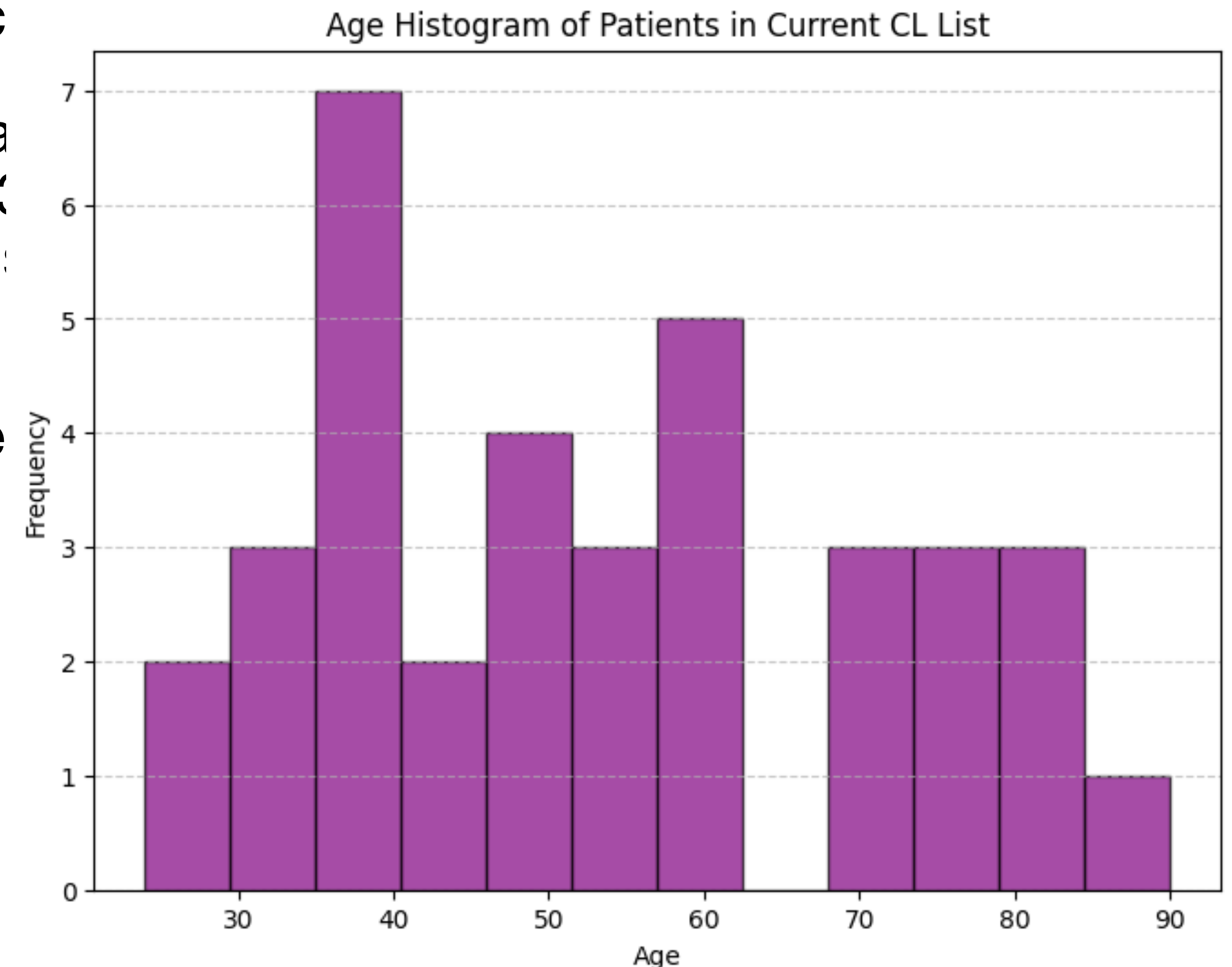
Why I picked this article?

- Has a very large sample size (UK biobank cohort >100,000 subjects)
- Addresses an age range that matches part of the demographics seen in the Psych CL service
- While the effects of alcohol on healthy individuals are of importance, most patients we see in the hospital have either health-related or socioeconomic risk factors

Study Demographics



- UK Biobank is a large-scale biomedical database and research resource containing de-identified genetic, lifestyle and health information and biologic
- From an initial pool of 217,462 UK Biobank participants, exclusions were made for incomplete data on alcohol consumption (non-drinkers (8,295), binge drinkers (20,167), early death (6,017))
- After exclusions, the final analytical sample consists of:
 - **Age:** 64.0 [62.0-67.0] years
 - **Biological Sex:** 67,693 [50.1%] women
 - **Race:** 94-99% white participants



Defining alcohol consumption and its patterns

- **Data Collection:** Participants reported the frequency and average amount of various alcoholic beverages consumed.
- **Alcohol Content Estimation:** Calculated by multiplying the volume consumed by the alcohol percentage (varied by beverage type) and the specific gravity of ethanol (0.789 g/mL) to get grams of alcohol per day.
- They used **occasional drinkers** (as opposed to non-drinkers) as their baseline group to account for *abstainer bias*

Category	Details
Mean Alcohol Intake	- Occasional: 2.86 g/day
	- Low Risk: Men: >2.86–20.00 g/day; Women: >2.86–10.00 g/day
	- Moderate Risk: Men: >20.00–40.00 g/day; Women: >10.00–20.00 g/day
	- High Risk: Men: >40.00 g/day; Women: >20.00 g/day
Alcohol Type Preference	- Preference for wine (>80% of alcohol intake from wine)
	- Preference for other drinks (beer, cider, spirits, etc.)
	- No preference (no dominant beverage type)
Drinking Context	- Drinkers only during meals
	- Drinkers either only outside meals or at any time
Combined Categories	- No wine preference nor drinking only during meals
	- Wine preference or drinking only during meals
	- Wine preference and drinking only during meals

Health Risk Factors - Frailty Index

- **Frailty Index (FI):** Health-related risk was assessed at baseline using the FI specifically developed for the UK Biobank.
- **Health Deficits Considered:** A total of 49 health deficits were evaluated. Some are scored dichotomously (0 vs. 1), others are scored based on severity (0, 0.25, 0.5, 1)
 - These include psychiatric symptoms too!
- **FI Calculation:**
 - Total points for all deficits were summed.
 - Divided by the number of deficits considered.
- **Health-Related Risk Classification:**
 - Participants were considered at risk if classified as prefrail or frail (FI > 0.12).

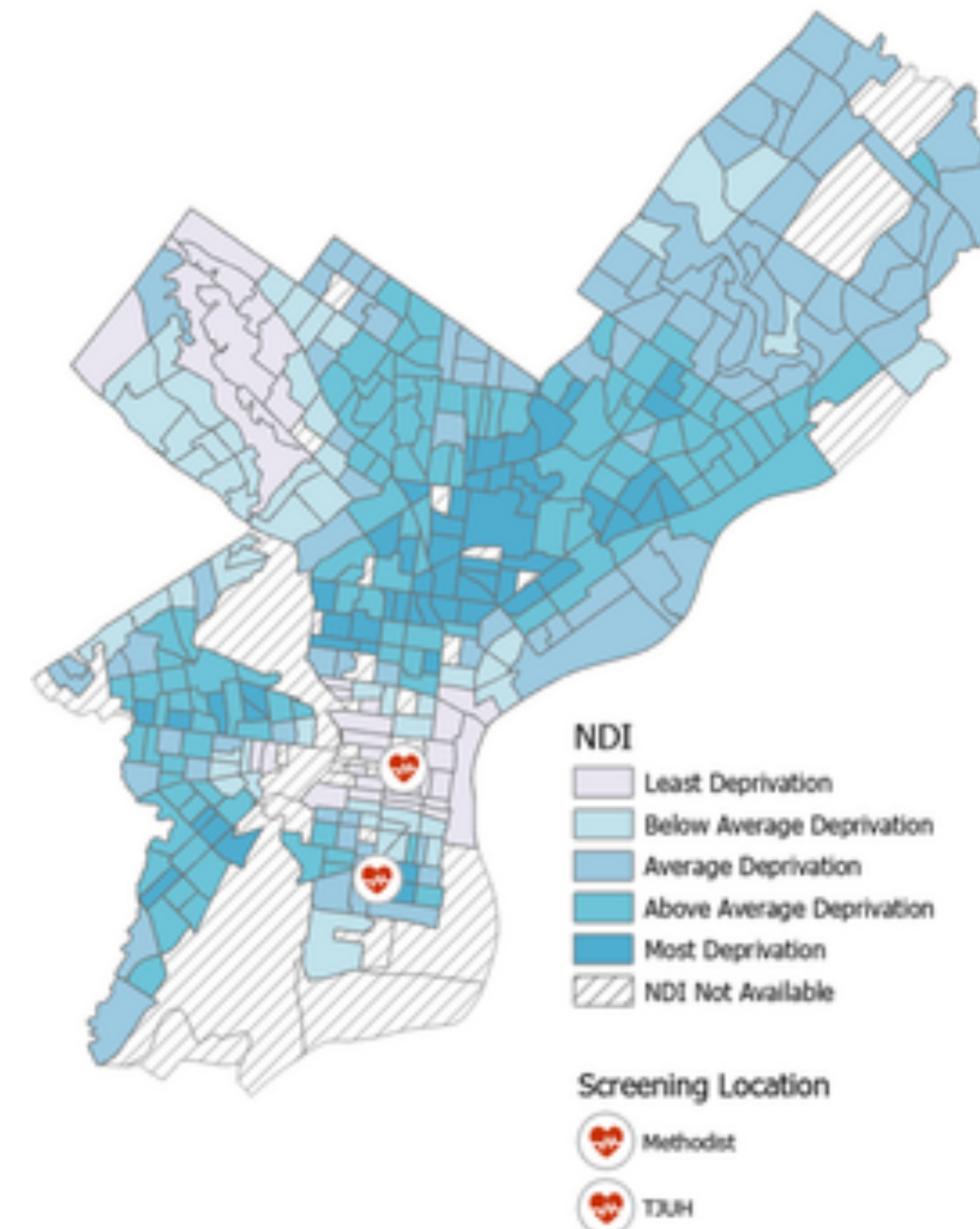
eTable 1. Health deficits of the Frailty Index in the UK Biobank cohort.

Health deficits	Scoring
1 Glaucoma *	No=0, Yes=1
2 Cataracts *	No=0, Yes=1
3 Hearing difficulty	No=0, Yes=1
4 Migraine *	No=0, Yes=1
5 Dental problems	No=0, Yes=1
6 Self-rated health	Excellent=0, Good=0.25, Fair=0.5, Poor=1
7 Fatigue: frequency of tiredness/lethargy in last 2 weeks	Not at all=0, Several days=0.25, More than half=0.5, Nearly every day=1
8 Sleep: experience of sleeplessness/insomnia	Never/rarely=0, Sometimes=0.5, Usually=1
9 Depressed feelings: frequency in last two weeks	Not at all=0, Several days=0.5, More than half=0.75, Nearly every day=1
10 Self-described nervous personality	No=0, Yes=1
11 Severe anxiety/panic attacks *	No=0, Yes=1
12 Common to feel loneliness	No=0, Yes=1
13 Sense of misery (ever/never)	No=0, Yes=1
14 Infirmary: long-standing illness or disability	No=0, Yes=1
15 Falls in last year	No falls=0, One fall=0.5, More than one=1
16 Fractures/broken bones in last five years	No=0, Yes=1
17 Diabetes *	No=0, Yes=1
18 Myocardial infarction *	No=0, Yes=1
19 Angina *	No=0, Yes=1
20 Stroke *	No=0, Yes=1
21 High blood pressure *	No=0, Yes=1
22 Hypothyroidism *	No=0, Yes=1
23 Deep-vein thrombosis *	No=0, Yes=1
24 High cholesterol *	No=0, Yes=1
25 Breathing: wheeze in last year	No=0, Yes=1
26 Pneumonia *	No=0, Yes=1
27 Chronic bronchitis/emphysema *	No=0, Yes=1
28 Asthma *	No=0, Yes=1
29 Rheumatoid arthritis *	No=0, Yes=1
30 Osteoarthritis *	No=0, Yes=1
31 Gout *	No=0, Yes=1
32 Osteoporosis *	No=0, Yes=1
33 Hayfever, allergic rhinitis or eczema *	No=0, Yes=1
34 Psoriasis *	No=0, Yes=1
35 Any cancer diagnosis *	No=0, Yes=1
36 Multiple cancers diagnosed	No cancer or single cancer=0, Multiple cancers=1
37 Chest pain	No=0, Yes=1
38 Head and/or neck pain	No=0, Yes=1
39 Back pain	No=0, Yes=1
40 Stomach/abdominal pain	No=0, Yes=1
41 Hip pain	No=0, Yes=1
42 Knee pain	No=0, Yes=1
43 Whole-body pain	No=0, Yes=1
44 Facial pain	No=0, Yes=1
45 Sciatica *	No=0, Yes=1
46 Gastric reflux *	No=0, Yes=1
47 Hiatus hernia *	No=0, Yes=1
48 Gall stones *	No=0, Yes=1
49 Diverticulitis *	No=0, Yes=1

*Self-reported medically diagnosed conditions.

Socioeconomic Risk Factors - Townsend Deprivation Index

- **Assessment Tool:** Socioeconomic risk was assessed at baseline using the Townsend Deprivation Index (TDI).
- **TDI Description:**
 - Measures the level of socioeconomic deprivation in an area.
 - Calculated using census data on variables such as unemployment, car ownership, home ownership, and household overcrowding.
- **TDI Range:** Scores ranged from –6.26 to 10.16, with higher scores indicating greater deprivation.
- **Risk Classification:** Participants were considered at socioeconomic risk if they lived in more deprived areas (TDI > 0).



Shusted et. al. 2023

Statistical Analysis

- **Primary Analysis:**

- Associations between alcohol consumption patterns (mean alcohol intake, wine preference, and drinking during meals) and all-cause and cause-specific mortality (cancer or CVD) were summarized using Cox regression models.
- Results presented as hazard ratios (HRs) with 95% confidence intervals (CIs).

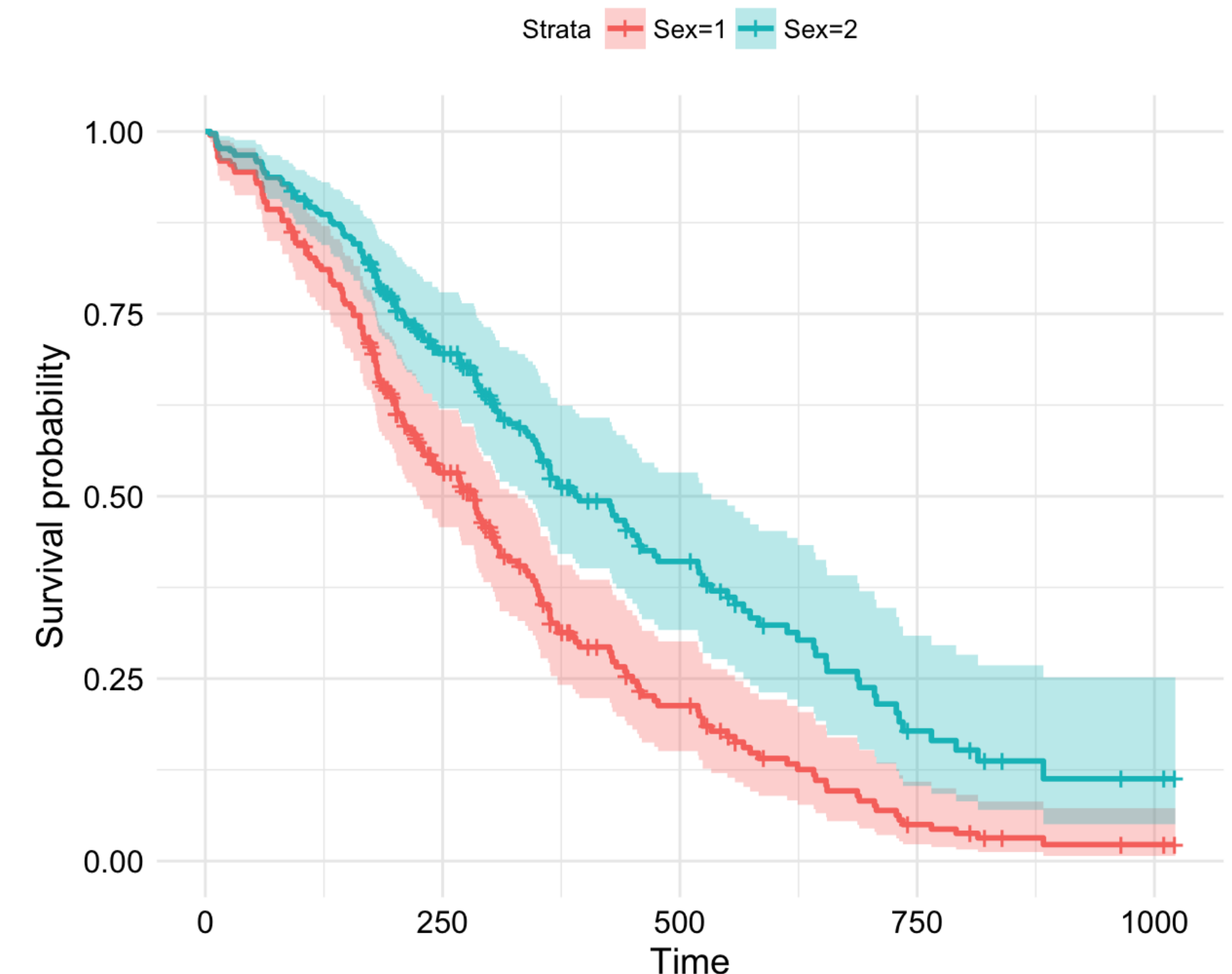
- **Adjustments in Cox Models:**

- *Adjusted for baseline sociodemographic factors:* sex, age, race/ethnicity, education, and TDI (Townsend Deprivation Index, excluded when stratifying by socioeconomic risk factors).
- *Adjusted for lifestyle factors:* tobacco smoking, leisure-time physical activity, and time spent watching television.
- *Adjusted for clinical characteristics:* diabetes, cardiovascular disease (CVD), cancer, and FI score (Frailty Index, excluded when stratifying by health-related risk factors).

- **Modification Analysis:**

- Evaluated whether wine preference and drinking during meals modified the relationship between mean alcohol intake and mortality using interaction terms based on combinations of drinking patterns:
 - No wine preference nor drinking only during meals.
 - Wine preference or drinking only during meals.
 - Wine preference and drinking only during meals.

Example of a Cox regression model



Associations between alcohol patterns and mortality

Less is always better!

Risk by Drinking Pattern (Compared to Occasional Drinkers):



Low-Risk Drinking:

- Higher cancer mortality: HR 1.11 (95% CI, 1.01–1.22).



Moderate-Risk Drinking:

- Higher all-cause mortality: HR 1.10 (95% CI, 1.03–1.18).
- Higher cancer mortality: HR 1.15 (95% CI, 1.05–1.27).



High-Risk Drinking:

- Higher all-cause mortality: HR 1.33 (95% CI, 1.24–1.42).
- Higher cancer mortality: HR 1.39 (95% CI, 1.26–1.53).
- Higher CVD mortality: HR 1.21 (95% CI, 1.04–1.41).

Health-Related and Socioeconomic Risks:

• Health-related risks:

- Increased all-cause and cancer mortality for moderate and high-risk drinkers with health-related risk factors.

• Socioeconomic risks:

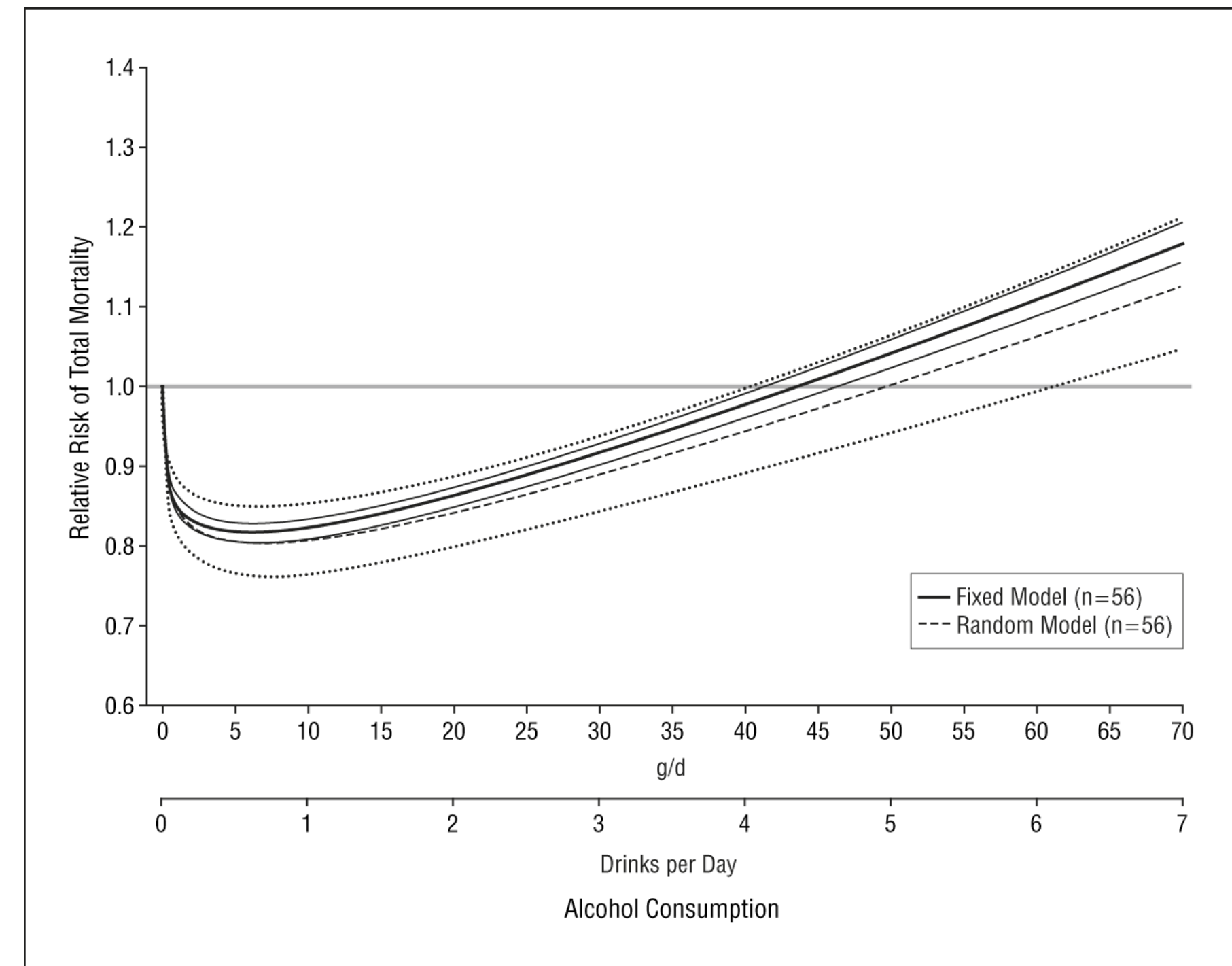
- Higher all-cause and cancer mortality for low, moderate, and high-risk drinkers with socioeconomic risk factors.

- Interestingly when the data was stratified like this, there was no increased mortality in individuals without health or socioeconomic risks except for high risk drinking (but there was also no benefit with lower risk drinking!)
- Cardiovascular mortality was only associated with high-risk drinking in this stratification

Why do these findings contradict prior literature?

- The authors claim that by using occasional drinkers as their control group they provide a “methodological innovation” that corrects for selection bias such as the abstainer bias
- That means, prior studies used abstainers, or non-drinkers, as their control population, whereas this study used occasional drinkers as their control population

Meta-analysis: Relative risk of total mortality as a function of alcohol consumption (~1M subjects)



Di Castelnuovo, JAMA Internal Medicine, 2006

Wine preference and drinking with meals 🍷 🍝

[I assume this was a required analysis because the authors are from Spain :)]

All-Cause Mortality

- **Participants with Health-Related Risk Factors:**
 - **Wine Preference:** HR 0.92 (95% CI, 0.87–0.97).
 - **Drinking Only During Meals:** HR 0.93 (95% CI, 0.89–0.97).
- **Participants with Socioeconomic Risk Factors:**
 - **Wine Preference:** HR 0.84 (95% CI, 0.78–0.90).
 - **Drinking Only During Meals:** HR 0.83 (95% CI, 0.78–0.89).

Cancer Mortality

- **Participants with Health-Related Risk Factors:**
 - **Drinking Only During Meals:** HR 0.92 (95% CI, 0.86–0.99).
- **Participants with Socioeconomic Risk Factors:**
 - **Drinking Only During Meals:** HR 0.85 (95% CI, 0.78–0.94).
 - **Wine Preference:** HR 0.89 (95% CI, 0.80–0.99).

CVD Mortality

Participants with Socioeconomic Risk Factors):

- **Drinking Only During Meals:** HR 0.86 (95% CI, 0.75–1.00).

Is drinking only wine and only with meals protective?

Table 4. Association of Mean Alcohol Intake Status With Mortality in Older Drinkers From the UK Biobank Cohort, by Drinking Patterns

Alcohol intake status ^a	All-cause mortality			Cancer mortality			CVD mortality		
	Deaths, No./total No.	HR (95% CI) ^{b,c}	P value for interaction	Deaths, No./total No.	HR (95% CI) ^{b,d}	P value for interaction	Deaths, No./total No.	HR (95% CI) ^{b,e}	P value for interaction
No wine preference nor drinking during meals									
Occasional	354/3354	1 [Reference]	NA	154/3353	1 [Reference]	NA	80/3353	1 [Reference]	NA
Low risk	2567/19 770	1.10 (0.98-1.23)	.89	1201/19 758	1.20 (1.01-1.42) ^f	.60	567/19 758	0.99 (0.78-1.26)	.43
Moderate risk	2461/16 948	1.17 (1.05-1.31) ^g	.36	1184/16 934	1.31 (1.10-1.55) ^g	.14	501/16 934	0.98 (0.77-1.24)	.80
High risk	2482/12 721	1.49 (1.33-1.67) ^h	.001	1167/12 715	1.63 (1.38-1.94) ^h	.02	551/12 715	1.35 (1.07-1.72) ^f	.007
Wine preference or drinking during meals									
Occasional	377/4054	1 [Reference]	NA	178/4052	1 [Reference]	NA	73/4052	1 [Reference]	NA
Low risk	2136/20 185	1.06 (0.95-1.18)	NA	1056/20 176	1.10 (0.94-1.30)	NA	448/20 176	1.08 (0.84-1.38)	NA
Moderate risk	1433/13 792	1.09 (0.97-1.22)	NA	741/13 786	1.15 (0.97-1.35)	NA	262/13 786	1.02 (0.79-1.33)	NA
High risk	896/7305	1.26 (1.11-1.42) ^h	NA	473/7301	1.32 (1.11-1.57) ^g	NA	169/7301	1.27 (0.97-1.68)	NA
Wine preference and drinking during meals									
Occasional	366/4641	1 [Reference]	NA	194/4640	1 [Reference]	NA	79/4640	1 [Reference]	NA
Low risk	1411/16 060	1.07 (0.95-1.20)	NA	755/16 054	1.07 (0.91-1.25)	NA	258/16 054	0.85 (0.66-1.10)	NA
Moderate risk	895/10934	1.04 (0.92-1.18)	NA	493/10 932	1.03 (0.87-1.22)	NA	163/10 932	0.90 (0.69-1.18)	NA
High risk	455/5339	1.06 (0.92-1.22)	NA	275/5337	1.13 (0.94-1.37)	NA	64/5337	0.72 (0.52-1.01)	NA

Study Limitations

- **Self-Reported Alcohol Intake:**
 - Alcohol consumption data were self-reported, introducing potential for misclassification.
- **Single Time-Point Measurement:**
 - Alcohol intake was measured only at baseline, without accounting for changes over time.
 - Unable to redistribute former drinkers among current drinker categories, potentially underestimating the true effects.
- **Residual Confounding:**
 - Despite adjustments for numerous confounders, residual confounding inherent to observational studies cannot be entirely ruled out.
- **Generalizability:**
 - Study population consisted of older adults in the UK with mostly White participants.
 - Results may not be applicable to other racial/ethnic groups, populations with different lifestyles, or those in differing socioeconomic contexts.

Summary of Findings

- **Low-Risk Drinkers:** Higher cancer mortality compared to occasional drinkers.
- **Moderate-Risk Drinkers:** Higher all-cause and cancer mortality.
- **High-Risk Drinkers:** Higher all-cause, cancer, and cardiovascular disease (CVD) mortality.
- Excess mortality greater in individuals with pre-existing medical conditions. Even low-risk drinkers had higher cancer mortality.
- Mortality hazard higher among individuals with socioeconomic disadvantages, where low alcohol consumption linked to increased all-cause and cancer mortality in this group.
- Drinking only with dinner and wine only blunted some of these associations, but did not really provide a benefit, just no damage